

T01 — Systems & OS Internals

Level 1 — Computing Foundations

1. Computer Architecture

Computer Architecture refers to the design and organization of a computer system and how its hardware and software components interact with each other. It defines the structure, functionality, and behavior of a computer system. Computer architecture acts as a blueprint that determines how data is processed, stored, and transmitted within a computer.

The primary purpose of computer architecture is to provide a framework that enables efficient execution of programs. It includes the design of the Central Processing Unit (CPU), memory systems, input/output devices, and communication pathways. A well-designed architecture improves performance, reliability, and scalability.

Computer architecture is generally divided into three categories: Instruction Set Architecture (ISA), Microarchitecture, and System Design. ISA defines the instructions that a processor can execute. Microarchitecture describes how the processor implements those instructions. System design focuses on integrating hardware components such as memory, storage, and peripherals.

One of the key concepts in computer architecture is the Von Neumann Architecture. In this architecture, instructions and data are stored in the same memory. The CPU fetches instructions from memory, decodes them, and executes them. Most modern computers are based on this architecture.

Another important aspect is the bus system. Buses are communication channels that transfer data between different components of a computer. The three major buses are the data bus, address bus, and control bus. Together they facilitate communication between CPU, memory, and I/O devices.

Computer architecture also includes performance optimization techniques such as pipelining, caching, and parallel processing. These techniques help improve processing speed and overall system efficiency. Modern processors use multiple cores and advanced architectures to handle large workloads efficiently.

Features

- Defines the structure and organization of a computer system.
- Includes CPU, memory, storage, and I/O devices.
- Provides communication between hardware components.

- Supports efficient execution of instructions.
- Uses architectures such as Von Neumann Architecture.
- Enables performance optimization through caching and pipelining.

Uses

- Designing computer hardware.
- Developing efficient software.
- Improving system performance.
- Building embedded systems and servers.
- Supporting modern computing devices.

Importance

- Forms the foundation of all computer systems.
- Helps achieve better speed and efficiency.
- Ensures compatibility between hardware and software.
- Enables technological advancements in computing.

2. CPU Fundamentals

The Central Processing Unit (CPU) is often called the brain of the computer because it performs calculations, executes instructions, and controls system operations. Every task performed by a computer ultimately passes through the CPU.

The CPU consists of three major components: the Arithmetic Logic Unit (ALU), the Control Unit (CU), and Registers. The ALU performs arithmetic and logical operations. The Control Unit manages the execution of instructions. Registers are small, high-speed storage locations used to hold data temporarily.

The CPU operates using the Fetch-Decode-Execute cycle. During the fetch phase, instructions are retrieved from memory. In the decode phase, the instruction is interpreted. Finally, during execution, the required operation is performed. This cycle repeats continuously while the computer is running.

Clock speed is an important characteristic of a CPU. It is measured in Hertz (Hz) and determines how many operations the processor can perform per second. Modern CPUs typically operate in Gigahertz (GHz), allowing billions of cycles per second.

Modern processors contain multiple cores. A single-core processor executes one task at a time, while multi-core processors can handle multiple tasks simultaneously. Dual-core, quad-core, and octa-core processors are common examples found in modern devices.

Cache memory is another essential component of CPU architecture. It stores frequently accessed data close to the processor, reducing access time. CPU caches are generally categorized into L1, L2, and L3 cache levels, with L1 being the fastest.

CPU performance depends on factors such as clock speed, number of cores, cache size, architecture design, and instruction execution efficiency. Together, these factors determine how effectively a processor can handle workloads.

Features

- Contains ALU, Control Unit, and Registers.
- Executes instructions using Fetch-Decode-Execute cycle.
- Supports multi-core processing.
- Includes cache memory for faster access.
- Performs arithmetic and logical operations.
- Controls overall system activities.

Uses

- Processing data and instructions.
- Running applications and operating systems.
- Managing system resources.
- Performing calculations.
- Supporting multitasking operations.

Importance

- Acts as the brain of the computer.
- Determines overall system performance.
- Enables execution of software programs.
- Supports complex computational tasks.
- Essential for all computing operations.

3. Memory Fundamentals

Memory is a crucial component of a computer system that stores data and instructions required for processing. It allows the CPU to access information quickly and efficiently during program execution.

Memory is broadly categorized into primary memory and secondary memory. Primary memory includes RAM and ROM, while secondary memory includes hard drives, SSDs, and optical storage devices. Primary memory is directly accessible by the CPU.

Random Access Memory (RAM) is volatile memory, meaning its contents are lost when power is turned off. RAM temporarily stores operating system files, applications, and active data. The more RAM a system has, the more programs it can handle simultaneously.

Read Only Memory (ROM) is non-volatile memory that retains data even when power is removed. ROM contains firmware and startup instructions required during the boot process. It plays an essential role in initializing hardware components.

Memory hierarchy is an important concept that organizes storage based on speed, cost, and capacity. Registers are the fastest but smallest storage units. Cache memory follows, then RAM, and finally secondary storage devices.

Virtual memory is a memory management technique that allows systems to use part of the storage device as additional RAM. It helps execute large programs even when physical memory is limited. However, virtual memory is slower than actual RAM.

Efficient memory management improves system performance, reduces processing delays, and enables multitasking. Modern operating systems use advanced memory allocation techniques to maximize resource utilization.

Features

- Stores data and instructions.
- Includes RAM, ROM, Cache, and Virtual Memory.
- Provides fast access to information.
- Organized in a memory hierarchy.
- Supports temporary and permanent storage.
- Enables multitasking.

Uses

- Holding active programs and data.
- Storing operating system files.
- Supporting application execution.
- Maintaining firmware instructions.
- Improving system responsiveness.

Importance

- Essential for smooth program execution.
- Directly affects system speed.
- Enables efficient multitasking.
- Supports data processing.
- Improves overall user experience.

4. Storage Systems

Storage systems are responsible for permanently storing data, applications, operating systems, and user files. Unlike RAM, storage devices retain information even when the computer is powered off.

Hard Disk Drives (HDDs) are traditional storage devices that use magnetic disks to store data. They offer large storage capacities at relatively low cost. However, their mechanical nature makes them slower compared to modern alternatives.

Solid State Drives (SSDs) use flash memory technology and contain no moving parts. They provide faster data access, quicker boot times, and improved reliability. SSDs have become the preferred storage solution in modern computers.

Storage devices are measured in units such as Kilobytes (KB), Megabytes (MB), Gigabytes (GB), Terabytes (TB), and Petabytes (PB). As data requirements grow, organizations increasingly adopt large-scale storage infrastructures.

Storage systems can be classified into Direct Attached Storage (DAS), Network Attached Storage (NAS), and Storage Area Networks (SAN). DAS connects directly to a computer, NAS provides file-based storage over a network, and SAN offers high-performance block-level storage.

Features

- Provides permanent data storage.

- Includes HDDs, SSDs, and cloud storage.
- Supports large storage capacities.
- Uses file organization techniques.
- Offers backup and recovery options.
- Can be local or network-based.

Uses

- Storing documents and media files.
- Saving operating systems and applications.
- Maintaining backups.
- Archiving business data.
- Supporting databases and servers.

Importance

- Prevents data loss after power shutdown.
- Enables long-term data retention.
- Supports business continuity.
- Provides reliable data access.
- Essential for digital information management.

5. Operating Systems

An Operating System (OS) is system software that acts as an interface between users and computer hardware. It manages hardware resources and provides services that allow applications to run efficiently.

The operating system performs several critical functions, including process management, memory management, file management, device management, and security management. These functions ensure smooth operation of the computer system.

Process management involves creating, scheduling, and terminating processes. The operating system allocates CPU time among multiple processes to achieve multitasking and efficient resource utilization.

Memory management is responsible for allocating and deallocating memory space to applications. The operating system ensures that programs do not interfere with each other's memory and efficiently manages available resources.

File management allows users and applications to create, modify, store, and retrieve files. The operating system organizes data into directories and maintains permissions to ensure security and accessibility.

Device management controls communication between hardware devices and software applications. Through device drivers, the operating system enables hardware components such as printers, keyboards, and storage devices to function correctly.

Examples of popular operating systems include Microsoft Windows, Linux, macOS, and Android. Operating systems provide the foundation upon which all software applications operate.

Features

- Manages hardware and software resources.
- Supports multitasking.
- Provides user interface.
- Handles memory management.
- Controls file systems and devices.
- Ensures security and access control.

Uses

- Running applications.
- Managing system resources.
- Controlling hardware devices.
- Providing user interaction.
- Supporting networking operations.

Importance

- Acts as a bridge between user and hardware.
- Makes computers easy to use.
- Ensures efficient resource allocation.
- Provides security and stability.
- Essential for system operation.

6. Processes & Threads

A process is a program that is currently being executed by the computer. When a user opens an application such as a web browser, text editor, or media player, the operating

system creates a process for that application. Each process has its own memory space, resources, and execution environment.

A process contains several components, including program code, data, stack, heap, and system resources. The code section contains instructions to be executed, while the stack stores function calls and local variables. The heap is used for dynamic memory allocation during program execution.

Processes can exist in different states during their lifecycle. Common process states include New, Ready, Running, Waiting (Blocked), and Terminated. The operating system continuously manages transitions between these states to ensure efficient utilization of CPU resources.

A thread is the smallest unit of execution within a process. A process may contain one or more threads. Threads within the same process share memory and resources, allowing them to communicate more efficiently than separate processes.

There are two types of threading models: Single-threaded and Multi-threaded. A single-threaded application performs one task at a time, while a multi-threaded application can execute multiple tasks simultaneously. Modern applications use multithreading to improve performance and responsiveness.

Processes and threads are fundamental to modern operating systems. They enable multitasking, parallel processing, and efficient resource utilization, allowing multiple applications and services to run concurrently on a computer system.

Features

- Processes have independent memory spaces.
- Threads share resources within a process.
- Supports parallel execution.
- Enables multitasking.
- Managed by operating systems.
- Uses scheduling mechanisms.

Uses

- Running multiple applications simultaneously.
- Improving application performance.
- Supporting background tasks.
- Handling multiple user requests.
- Performing parallel computations.

Importance

- Increases system efficiency.
- Enhances responsiveness.
- Optimizes CPU utilization.
- Supports modern software development.
- Enables concurrent execution.

7. File Systems

A file system is a method used by an operating system to organize, store, and retrieve data on storage devices. It provides a logical structure that allows users and applications to access files efficiently.

Files are organized into directories or folders. A directory can contain files as well as subdirectories, creating a hierarchical structure. This organization helps users manage and locate data more effectively.

The file system maintains metadata for every file. Metadata includes information such as file name, size, creation date, modification date, permissions, and storage location. This information helps the operating system manage files efficiently.

Different operating systems use different file systems. Examples include FAT32, NTFS, exFAT, and ext4. NTFS is commonly used in Windows systems, while ext4 is widely used in Linux environments. Each file system offers unique features and capabilities.

File permissions play an important role in security. Permissions determine who can read, write, modify, or execute files. Operating systems use permission mechanisms to prevent unauthorized access and protect sensitive information.

Journaling is a feature available in many modern file systems. Before making changes, the file system records the intended operation in a journal. If a system crash occurs, the journal helps recover data and maintain consistency.

Efficient file systems improve storage performance, data reliability, and security. They form a critical part of operating systems and enable effective management of digital information across various storage devices.

Features

- Organizes data into files and directories.
- Maintains file metadata.

- Supports permissions and security.
- Provides data retrieval mechanisms.
- Supports journaling.
- Works with different storage devices.

Uses

- Managing files and folders.
- Controlling data access.
- Storing user information.
- Organizing operating system files.
- Supporting application data storage.

Importance

- Enables efficient data organization.
- Improves data security.
- Facilitates data recovery.
- Simplifies file management.
- Essential for storage systems.

8. System Boot Process

The system boot process refers to the sequence of steps performed when a computer is powered on. Its purpose is to initialize hardware components and load the operating system into memory.

When the power button is pressed, the CPU begins executing firmware instructions stored in ROM. Modern systems use BIOS (Basic Input/Output System) or UEFI (Unified Extensible Firmware Interface) firmware to start the initialization process.

The firmware performs a Power-On Self-Test (POST). During POST, hardware components such as memory, processor, keyboard, and storage devices are checked to ensure they are functioning correctly. If a problem is detected, error messages or beep codes may be generated.

After successful hardware verification, the firmware locates a bootable storage device. It then loads the bootloader, a small program responsible for loading the operating system kernel into memory.

The operating system kernel is the core component of the OS. Once loaded, it initializes memory management, process management, device drivers, and other essential system services required for operation.

After kernel initialization, system services and background processes are started. These services provide networking, security, printing, and other functionalities required by the operating system.

Finally, the user interface is displayed, allowing users to log in and interact with the computer. The boot process ensures that all hardware and software components are properly initialized before the system becomes operational.

Features

- Starts when the computer powers on.
- Performs hardware initialization.
- Executes POST checks.
- Loads bootloader and operating system.
- Initializes system services.
- Prepares the user environment.

Uses

- Starting computer systems.
- Verifying hardware functionality.
- Loading operating systems.
- Initializing drivers and services.
- Preparing applications for use.

Importance

- Ensures proper system startup.
- Detects hardware issues.
- Loads essential software components.
- Enables system functionality.
- Critical for computer operation.

9. Applications & Services

Applications are software programs designed to perform specific tasks for users. Examples include word processors, web browsers, media players, and database

management systems. Applications rely on the operating system for access to hardware resources.

Applications can be classified into system applications and user applications. System applications assist in managing computer operations, while user applications help perform everyday tasks such as document editing, communication, and entertainment.

A service is a background program that runs without direct user interaction. Services provide essential functionalities required by the operating system and other applications. They start automatically when the system boots.

Examples of services include web servers, database servers, print spoolers, antivirus engines, and network management services. These services continuously monitor and perform specific tasks in the background.

Operating systems use service managers to control services. Service managers can start, stop, restart, and monitor services as needed. This helps ensure system stability and efficient resource management.

Applications often depend on multiple services to function correctly. For example, a web application may require a database service, networking service, and authentication service. These components work together to deliver functionality to users.

Applications and services form the software ecosystem of a computer system. They enable users to perform tasks while background services ensure smooth operation, security, and communication between software and hardware components.

Features

- Applications provide user functionality.
- Services run in the background.
- Support task automation.
- Depend on operating system resources.
- Can communicate over networks.
- Support various business functions.

Uses

- Document editing.
- Internet browsing.
- Database management.
- Communication and collaboration.

- Background system operations.

Importance

- Enables productive computing.
- Supports business processes.
- Provides user-friendly functionality.
- Maintains system operations.
- Improves automation and efficiency.

10. Virtualization Basics

Virtualization is a technology that allows multiple virtual machines (VMs) to run on a single physical computer. It enables efficient utilization of hardware resources by sharing CPU, memory, storage, and networking among multiple virtual environments.

A virtual machine is a software-based computer that behaves like a physical computer. Each VM has its own operating system, applications, and virtual hardware. Users can run multiple operating systems simultaneously on the same physical machine.

The software responsible for managing virtual machines is called a hypervisor. A hypervisor creates and controls virtual environments while allocating resources to each virtual machine. It acts as a layer between hardware and virtual machines.

There are two types of hypervisors. Type 1 hypervisors run directly on hardware and offer better performance. Examples include VMware ESXi and Microsoft Hyper-V. Type 2 hypervisors run on top of an operating system, such as Oracle VM VirtualBox.

Virtualization offers several benefits, including better hardware utilization, reduced infrastructure costs, simplified management, and improved disaster recovery. Organizations can run multiple servers on a single physical machine, reducing hardware expenses.

Virtualization also supports testing and development environments. Developers can create isolated virtual machines to test software without affecting the host system. This improves flexibility and reduces deployment risks.

Today, virtualization serves as the foundation for cloud computing platforms. It enables cloud providers to efficiently allocate resources, scale services, and deliver computing infrastructure to users on demand. Virtualization has become an essential technology in modern IT environments.

Features

- Creates virtual machines.
- Uses hypervisors.
- Supports multiple operating systems.
- Provides isolated environments.
- Shares physical resources.
- Enables cloud computing.

Uses

- Server consolidation.
- Software testing.
- Development environments.
- Disaster recovery.
- Cloud infrastructure deployment.

Importance

- Reduces hardware costs.
- Improves resource utilization.
- Enhances scalability.
- Simplifies system management.
- Forms the foundation of modern cloud computing.